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Centro Nacional de Supercomputación



Providentia

An Evaluation and Visualisation Tool for ACTRIS Data

05/06/2025

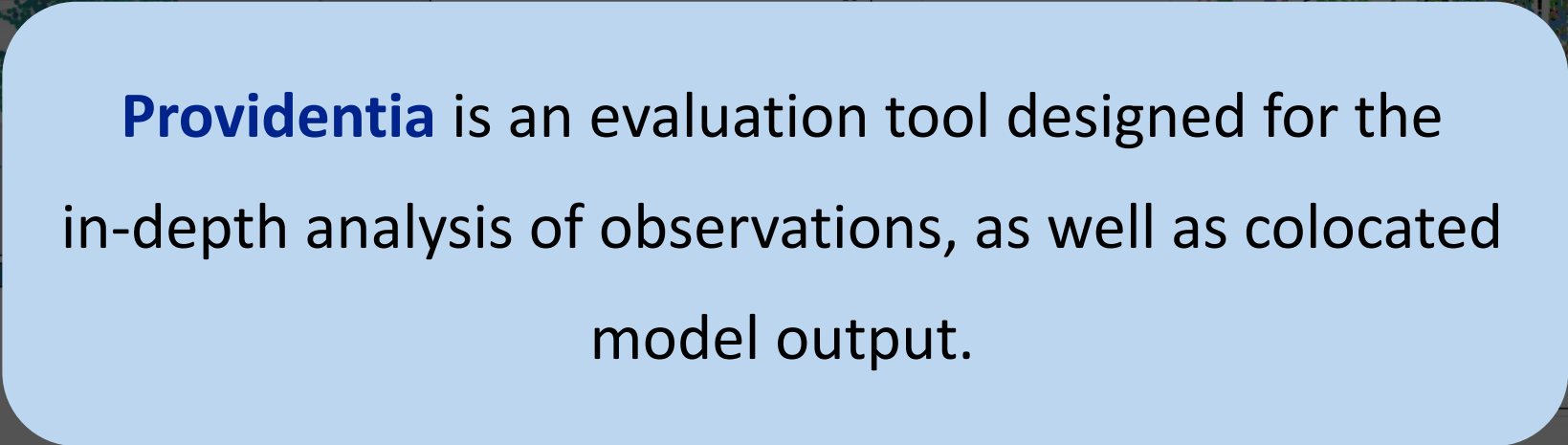
Paula Serrano | Alba Vilanova | Dene Bowdalo

Introduction



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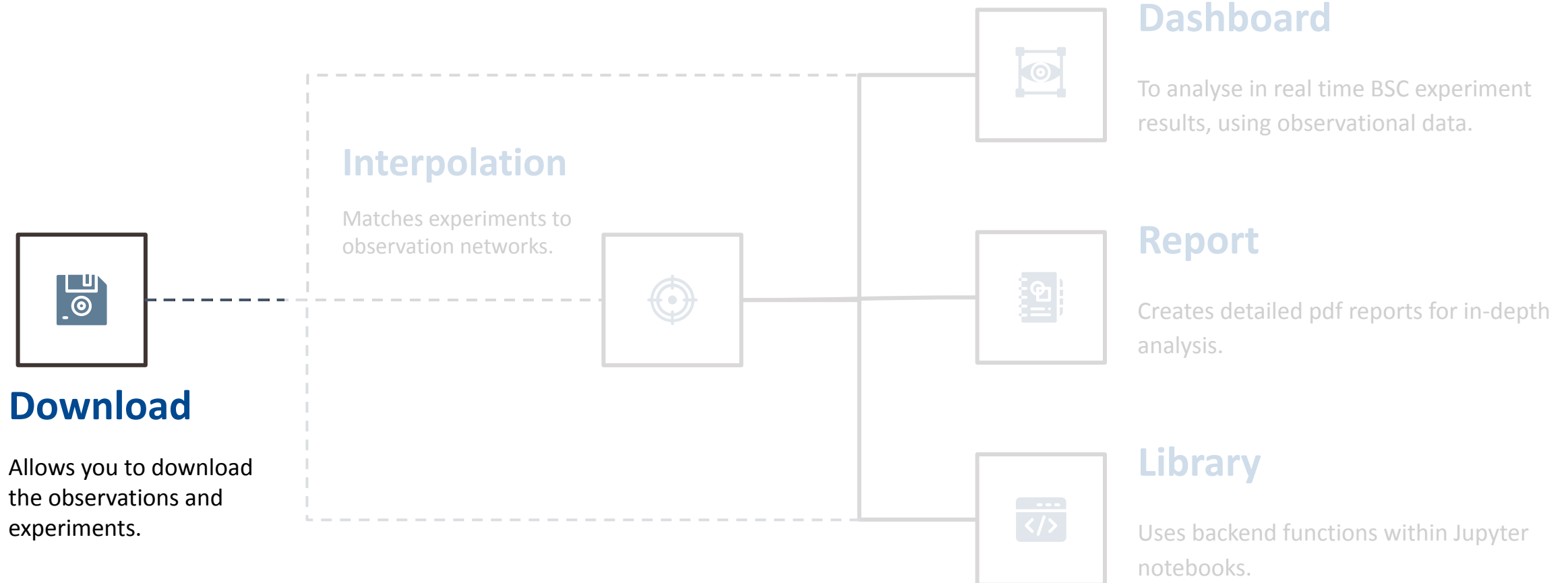
Key Functionalities



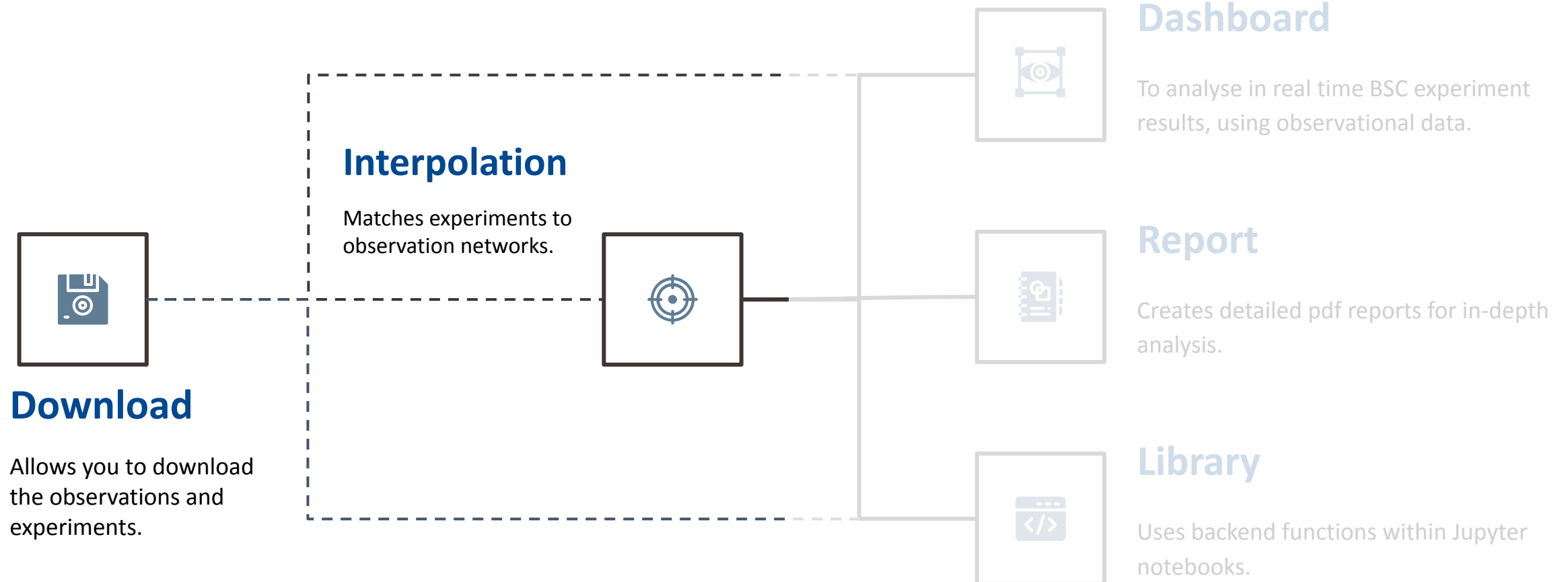
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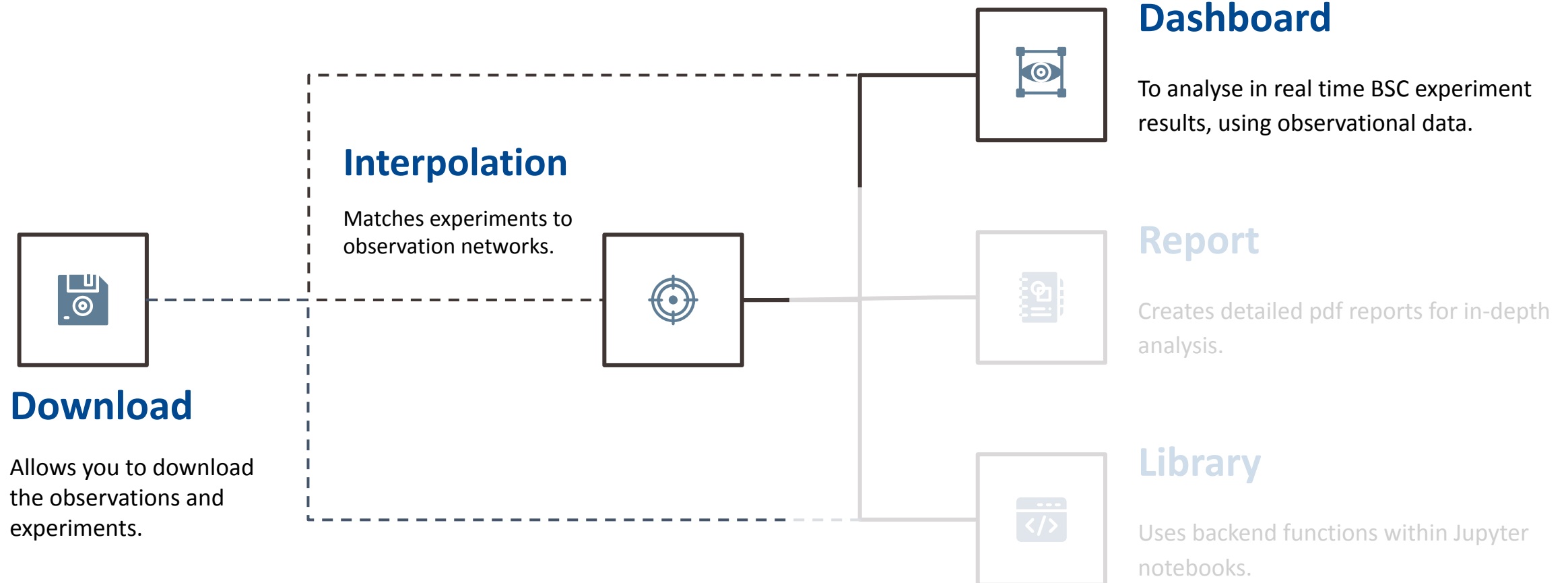
Structure



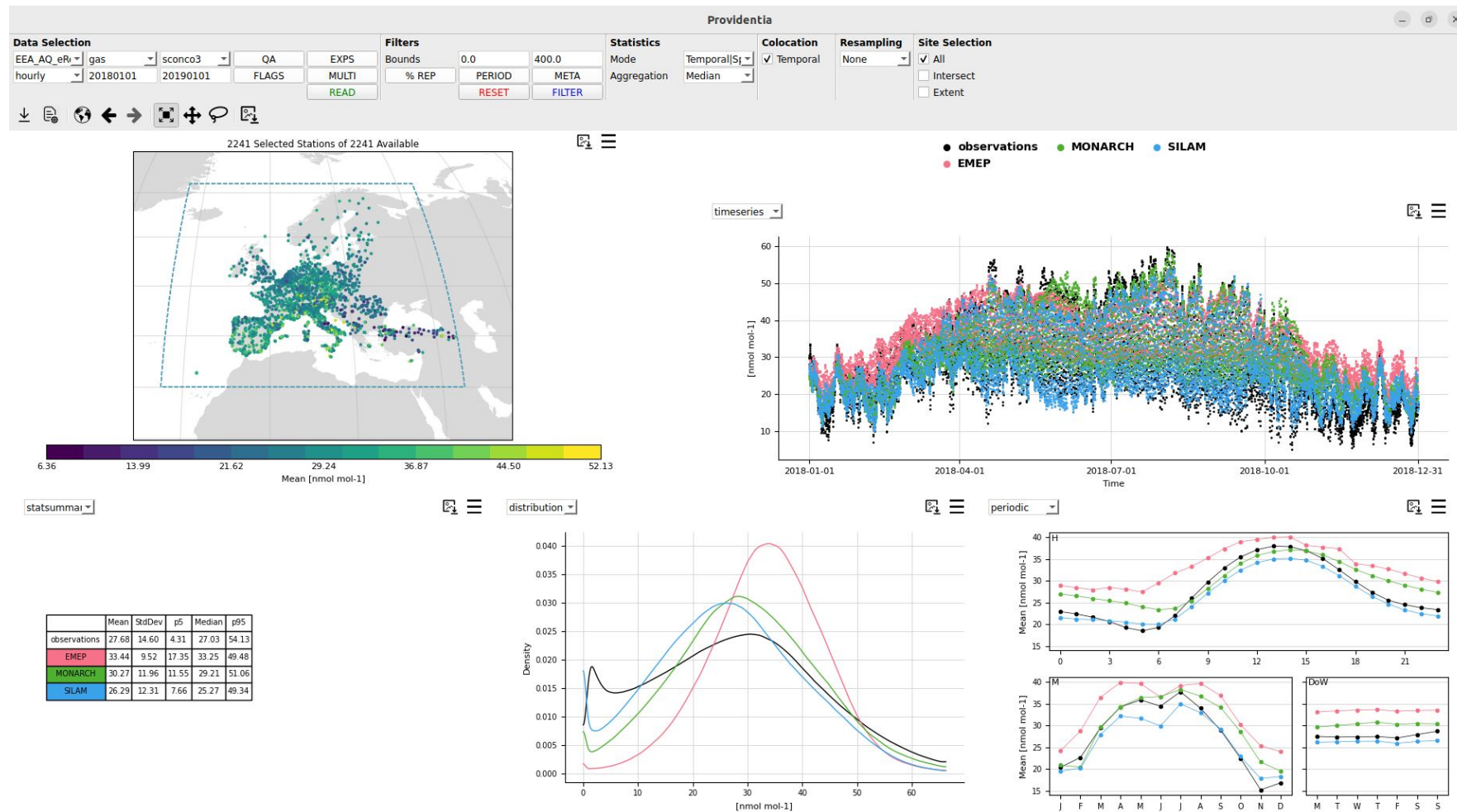
Structure



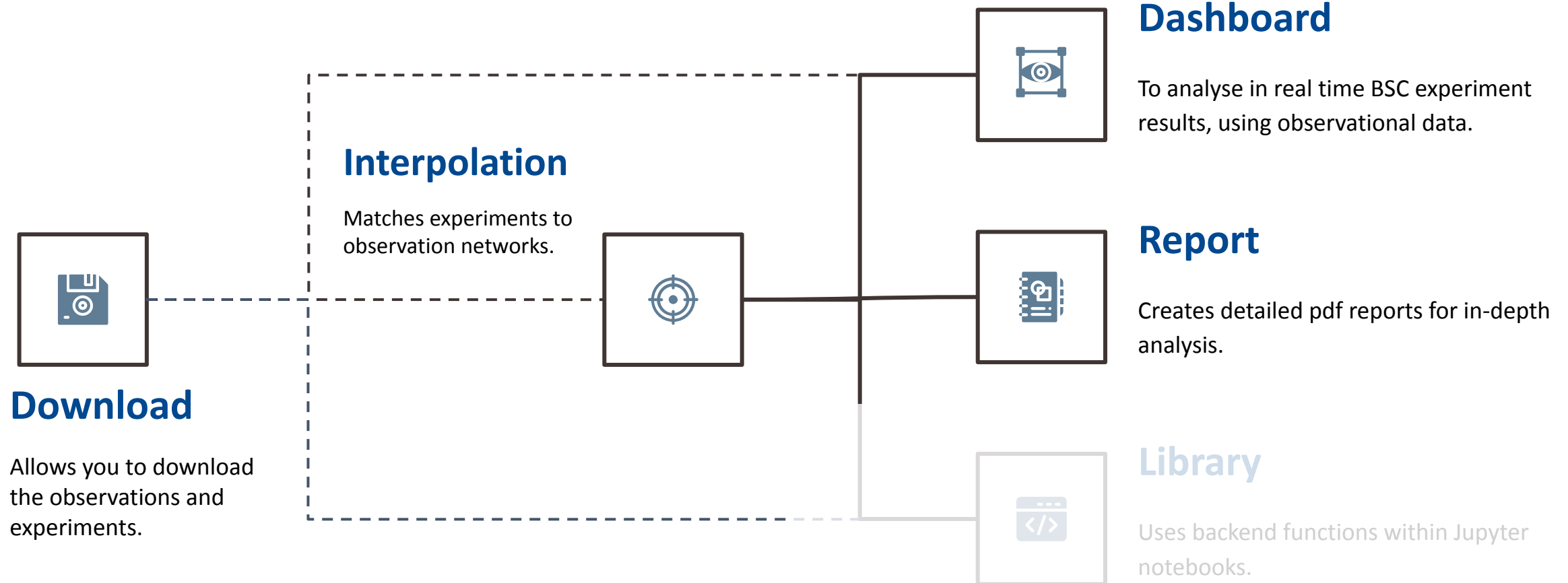
Structure



Dashboard



Structure



Report

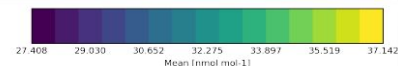
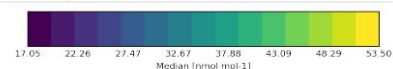
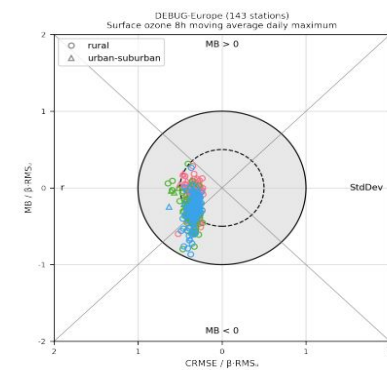


Provident

```

Network :
Species : [
Temporal Reso
Date Range : 2018
Experiments : ['EMEP'
Temporal Colc
Spatial Coloc
Subsections : ['DEBUG'

```

Map Median (Summary)
EBAS|sconco3Violin (Summary)
EBAS|sconco3Timeseries (Summary)
EBAS|sconco3Fairmode target (Summary)
EBAS|sconco3
$$\begin{aligned}\alpha &= 0.79 \\ \beta &= 2 \\ RV &= 60.12 \\ U_{\text{H}_2\text{O}, RV} &= 0.18\end{aligned}$$

134 stations with
coverage above 75%

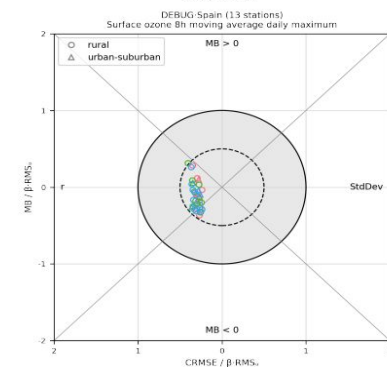
EMEP
MOI₉₀ = 0.50

0 stations with MQI > 1

MONARCH
MQI₉₀ = 0.60

SILAM

$MQI_{90} = 0.66$
0 stations with $MQI > 1$


$$\begin{aligned}\alpha &= 0.79 \\ \beta &= 2 \\ RV &= 60.12 \\ U_{20,r}^{RV} &= 0.18\end{aligned}$$

13 stations with
coverage above 75%

EMEP
MOI_{max} = 0.41

0 stations with MQI > 1

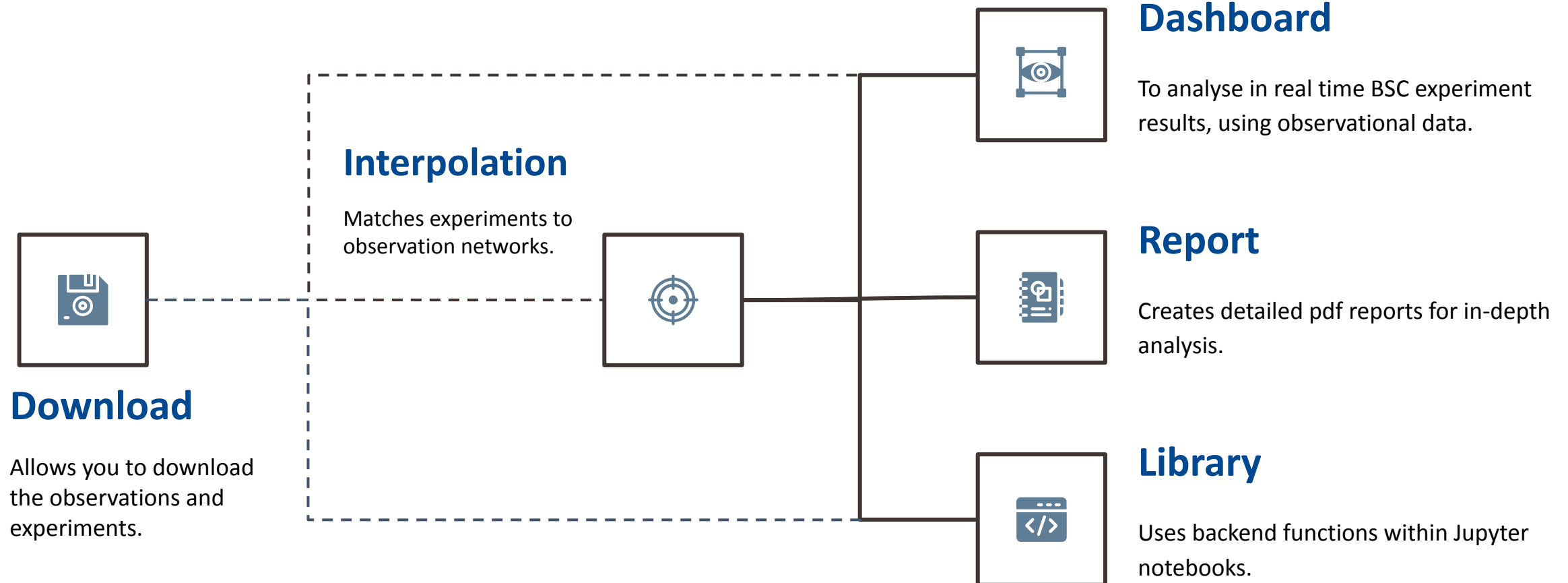
MONARCH
MQI₉₀ = 0.43

0 stations with MQI > 1

511 AM

MQI₉₀ = 0.44
0 stations with MQI > 1

Structure



Library

```
import providentia as prv
```

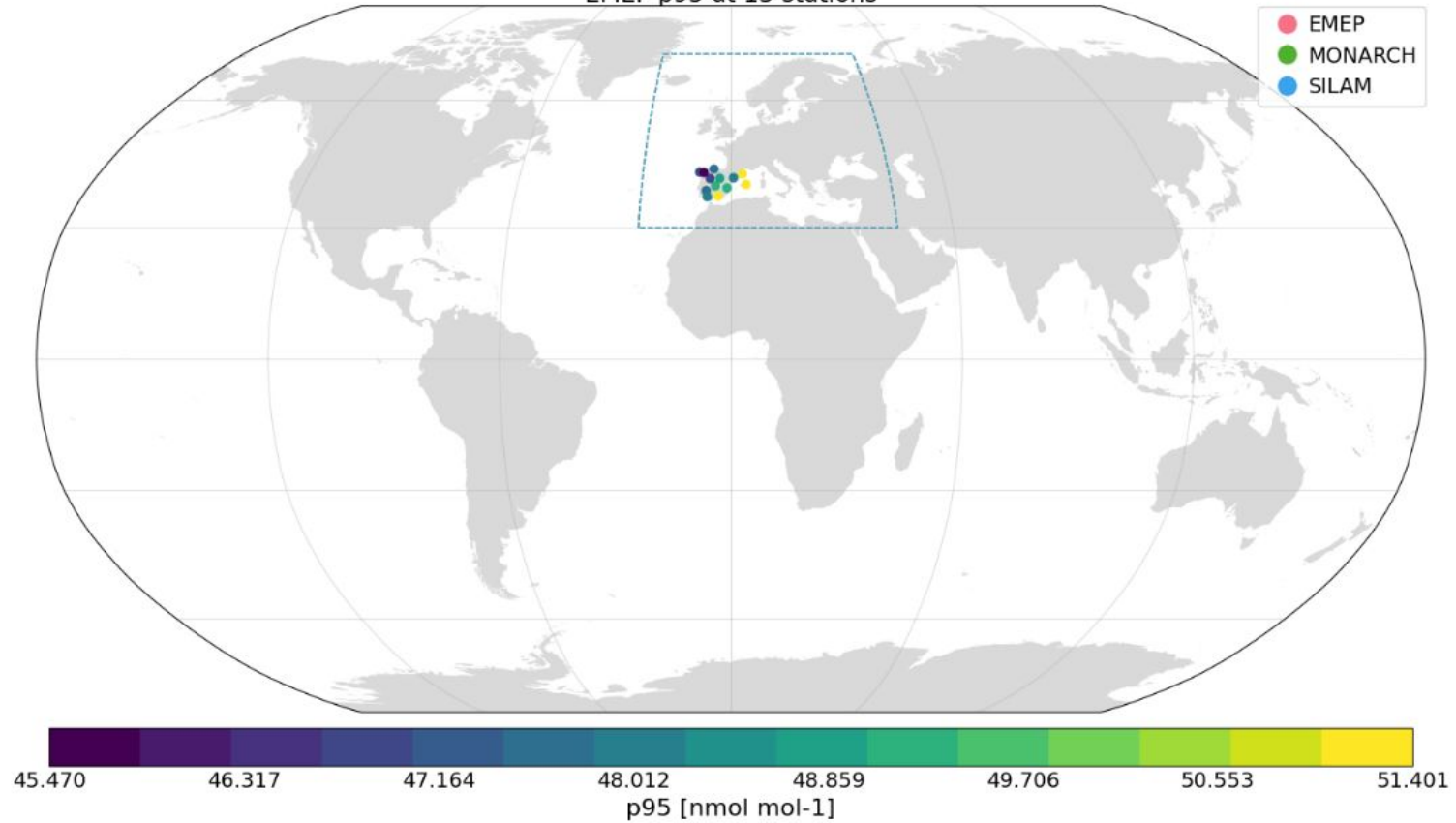
```
provi = prv.load("interactive_template.conf")
```

```
: # make a map plot (p95 statistic)  
provi.make_plot('map-p95', labela='EMEP', map_extent=[-180, 180, -90, 90], plot_options=['domain'])
```

Warning: Width and/or height have not been passed. The default values will be set.

Warning: More than 1 network or species defined, can only plot for 1 pair. Taking EBAS|sconco3.

EMEP p95 at 13 stations



ACTRIS



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ACTRIS Configuration Setup

To use ACTRIS data in Providentia, you just need to set up a **configuration file**.

configurations/actris.conf

Section ([name])

```
[ACTRIS]
framework = actris
species = sconco3
resolution = hourly
start_date = 20180101
end_date = 20190101
experiments = cams61_chimere_ph2-eu-000 (CHIMERE)
temporal_colocation = True
spatial_colocation = True
report_type = standard
report_summary = True
report_stations = False
report_filename = PROVIDENTIA_Report
report_title = Report
```

Subsections ([[name]])

```
[[Barcelona]]
latitude = 39.8, 41.8
longitude = 1.5, 2.5
```

```
[[Spain]]
longitude = -9.39, 3.04
latitude = 35.945, 43.75
```



Download ACTRIS data

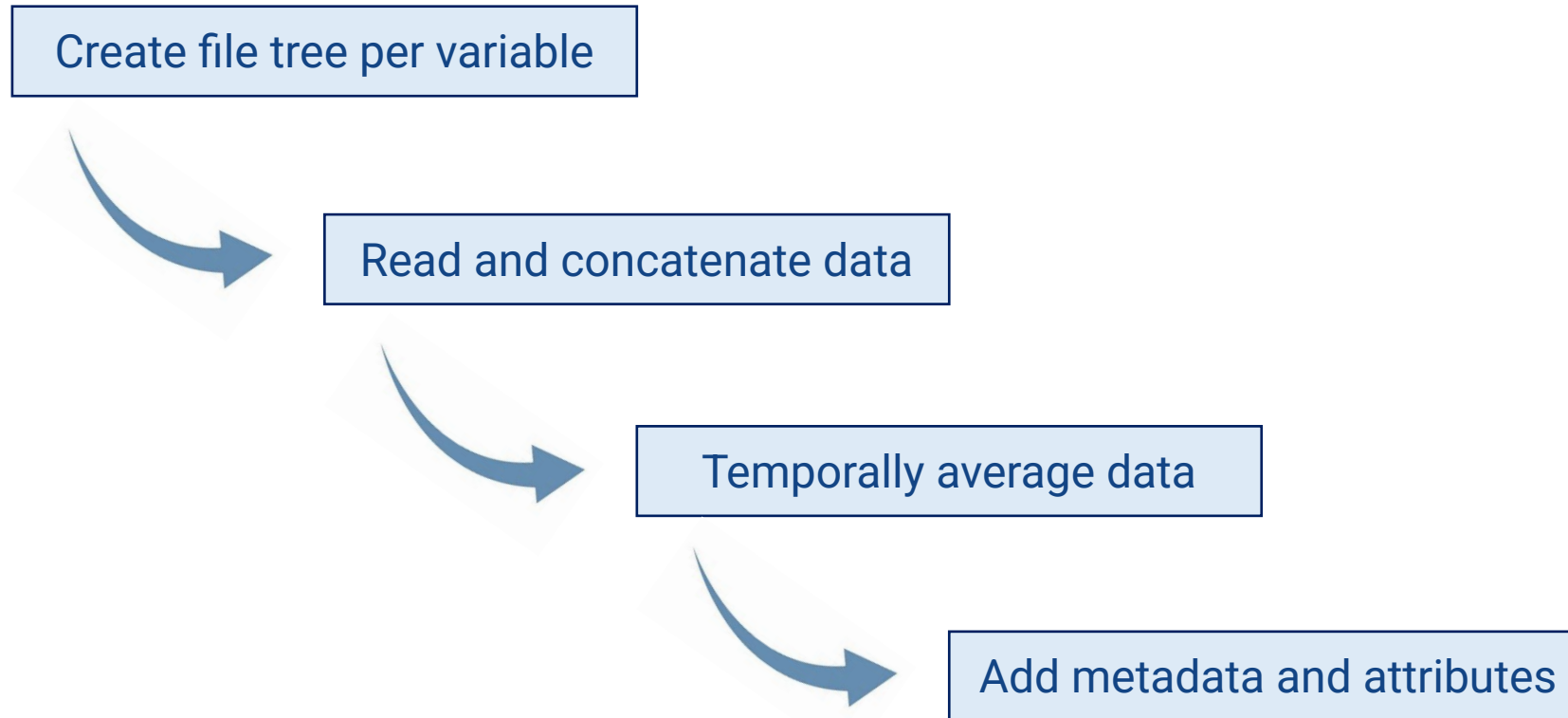
Using the download mode from Providentia, we can access Thredds (EBAS DOI in prod-actris-md.nilu.no) and retrieve the files to our machine in BSC format for the selected **variable, resolution and period**.

Once the config file is ready, downloading is simple. You run a single command:

```
./bin/providentia --config=your_configuration_name.conf --download
```

Providentia will automatically fetch the relevant ACTRIS data and begin formatting it.

Data Formatting



Demo



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Launch the tool

- Opening the dashboard:

```
./bin/providentia --config='configuration_name.conf'
```

- Opening the jupyter notebook:

```
./bin/providentia --config='configuration_name.conf' --notebook
```

- Creating reports:

```
./bin/providentia --config='configuration_name.conf' --offline
```

- Interpolate the experiment to the observation:

```
./bin/providentia --config='configuration_name.conf' --interpolation
```



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Thank you for your attention!

paula.serrano@bsc.es | alba.vilanova@bsc.es | dene.bowdalo@bsc.es

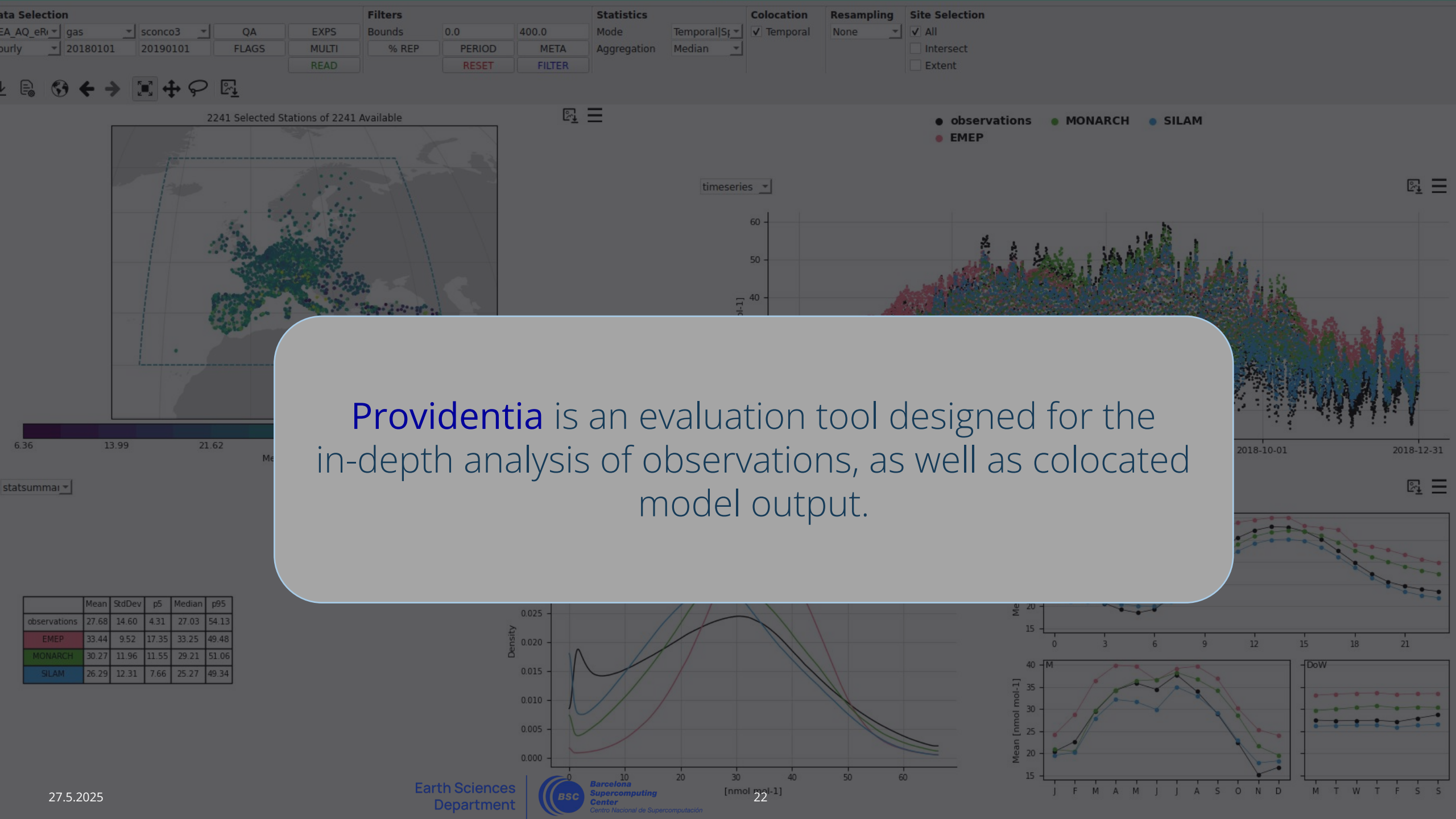


Providentia

An Evaluation and Visualisation Tool for ACTRIS Data



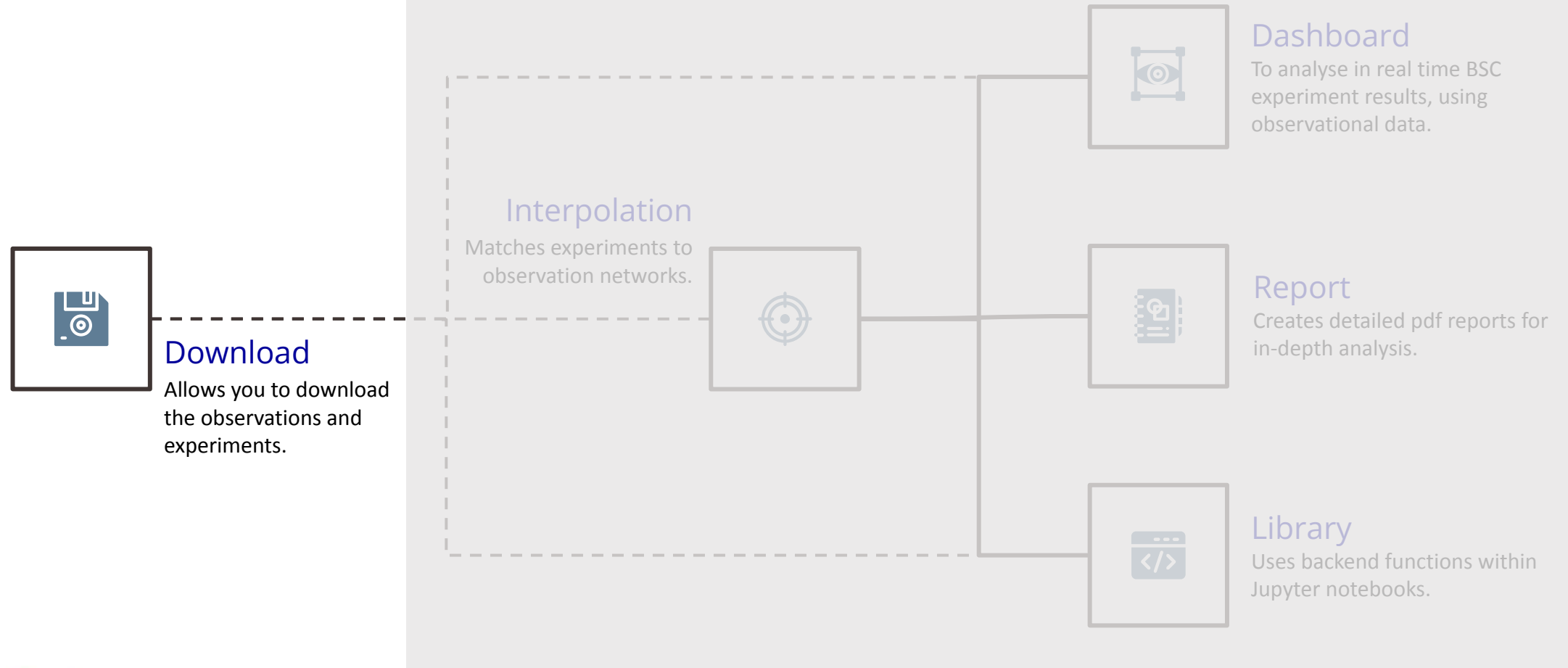
Introduction



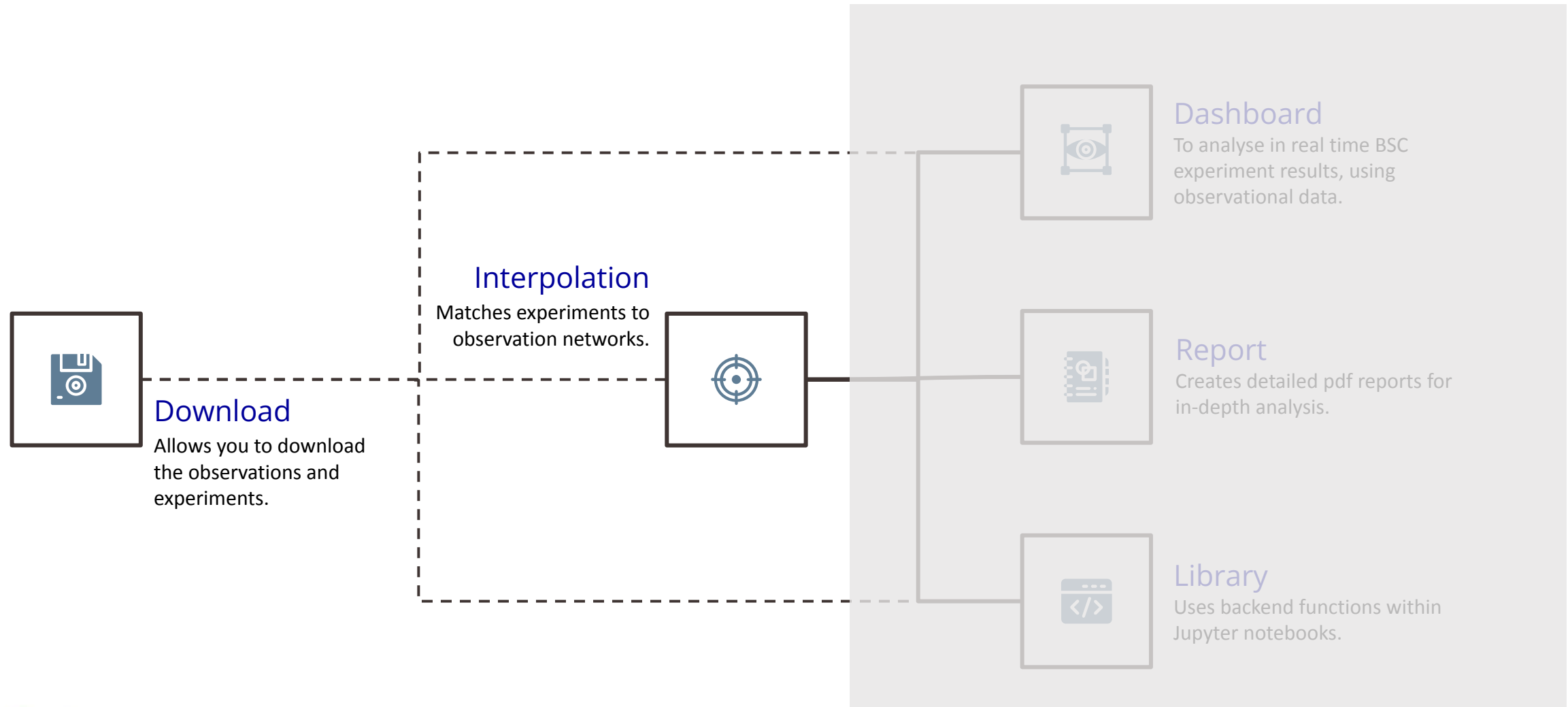
Providentia is an evaluation tool designed for the in-depth analysis of observations, as well as colocated model output.

Key Functionalities

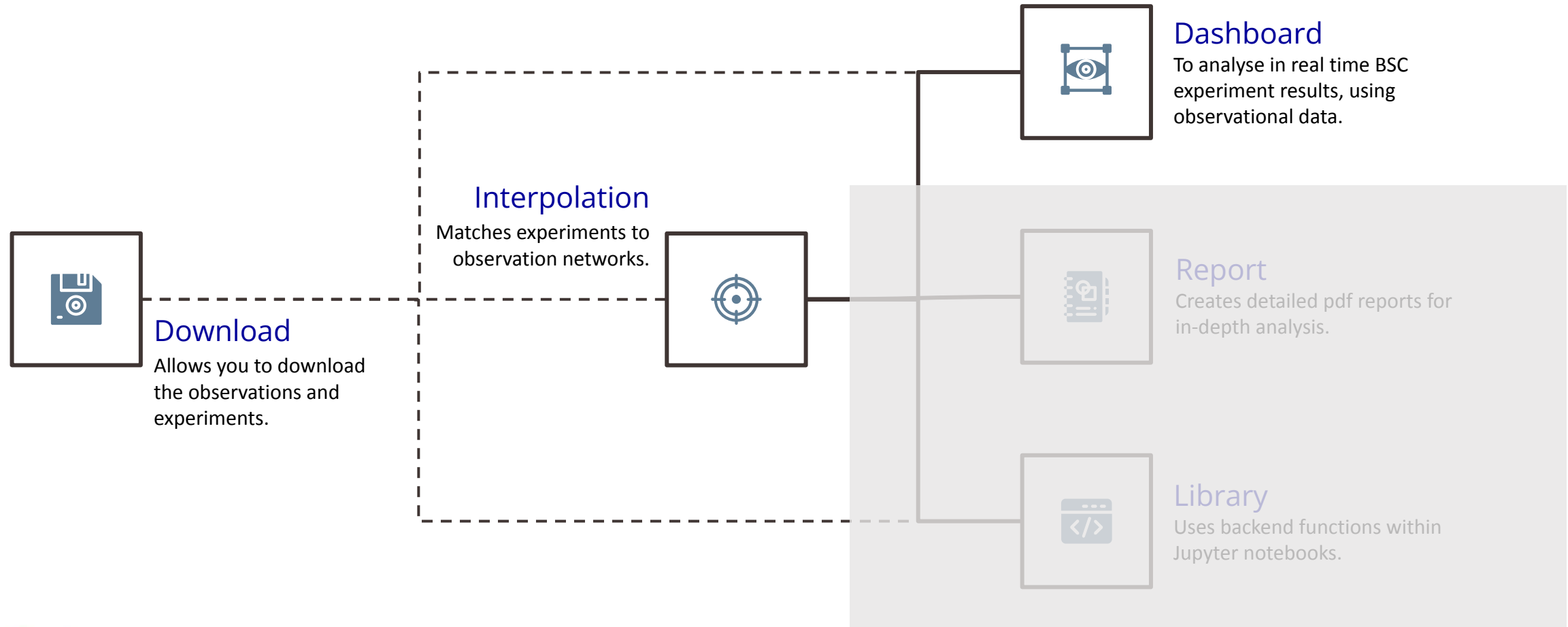
Structure



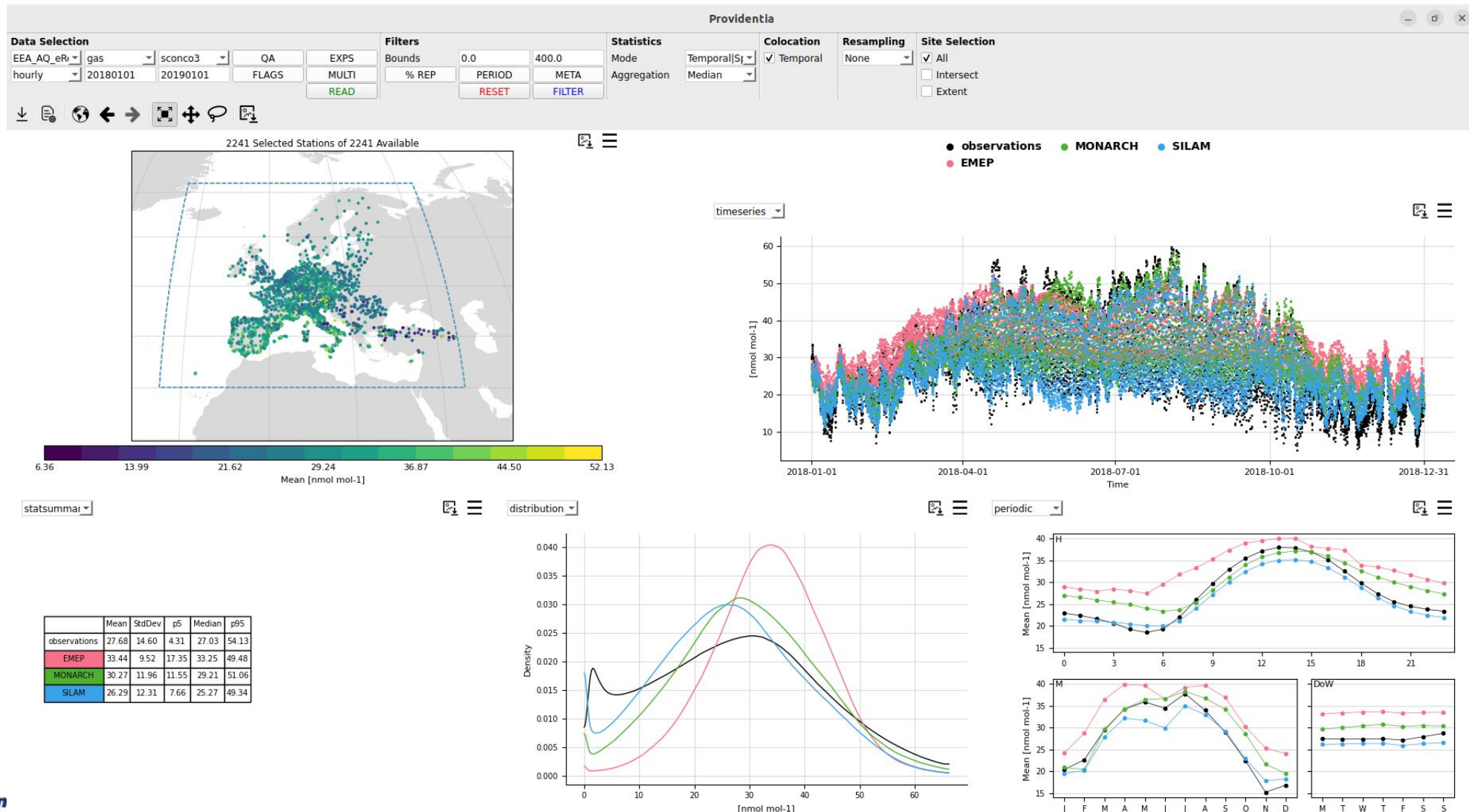
Structure



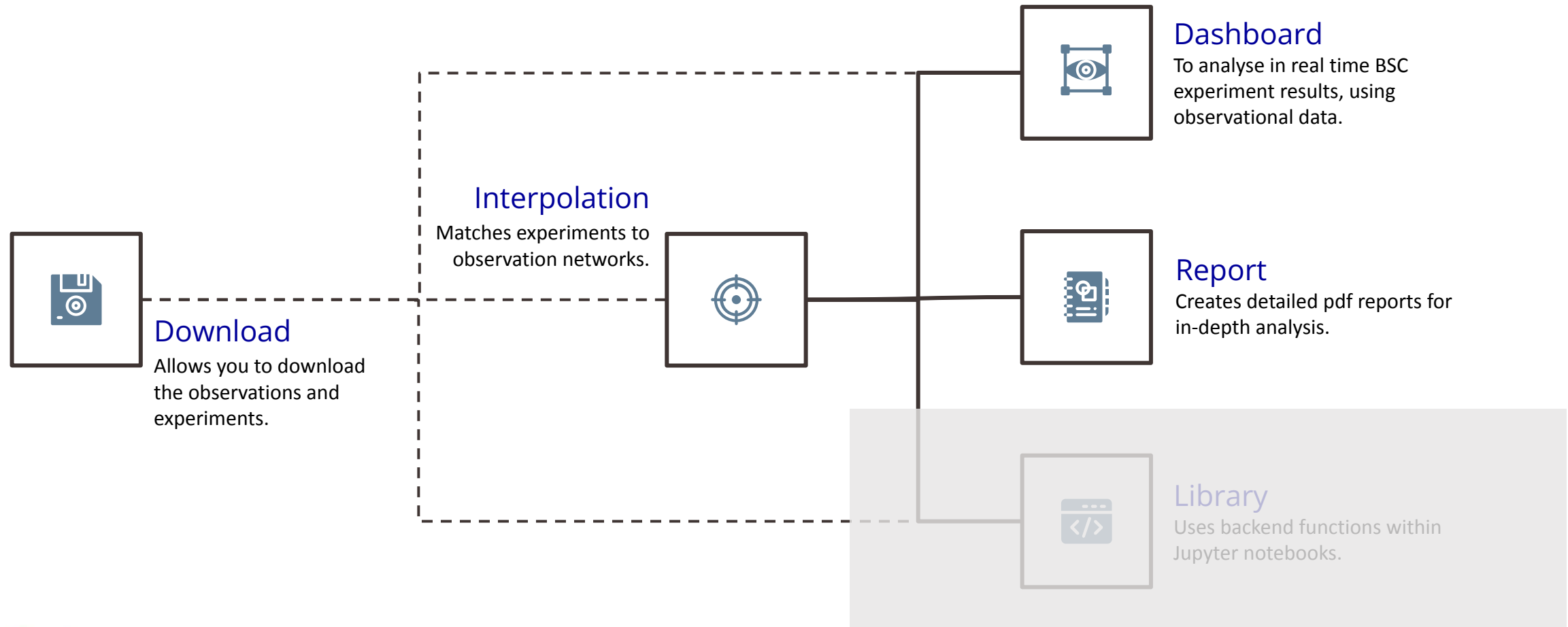
Structure



Dashboard



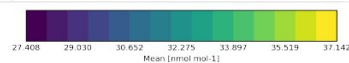
Structure



Report



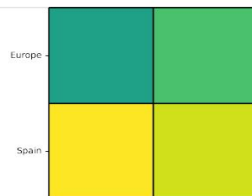
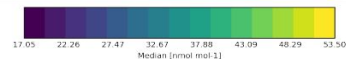
Heatmap Mean (Summary)
multispecies



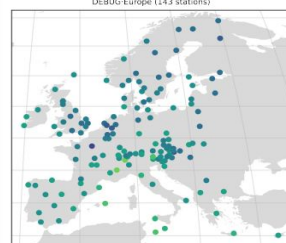
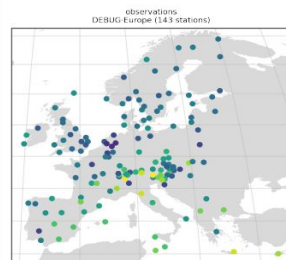
Providentia R

Network : ['EB/']
Species : ['scon']
Temporal Resolution :
Date Range : 20180101
Experiments : ['EMEP', 'MONARCH', 'SILAM']
Temporal Colocation :
Spatial Colocation :
Subsections : ['DEBUG-Europe']

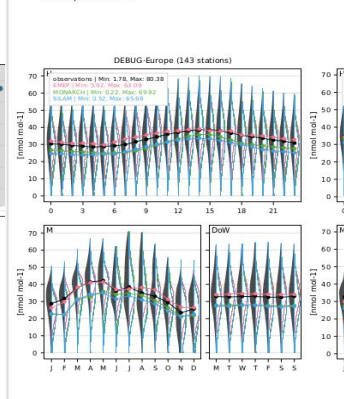
Map Median (Summary)
EBAS[sconco3]



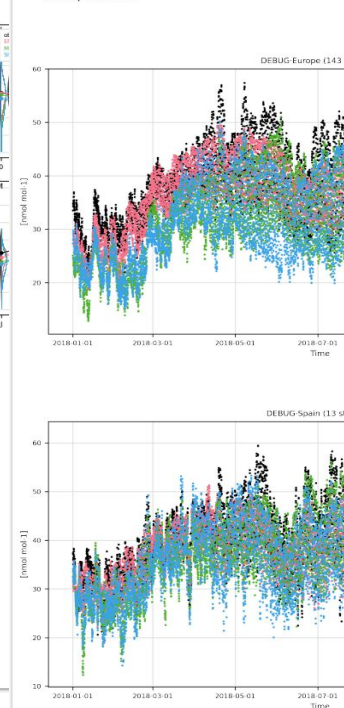
observations EMEP



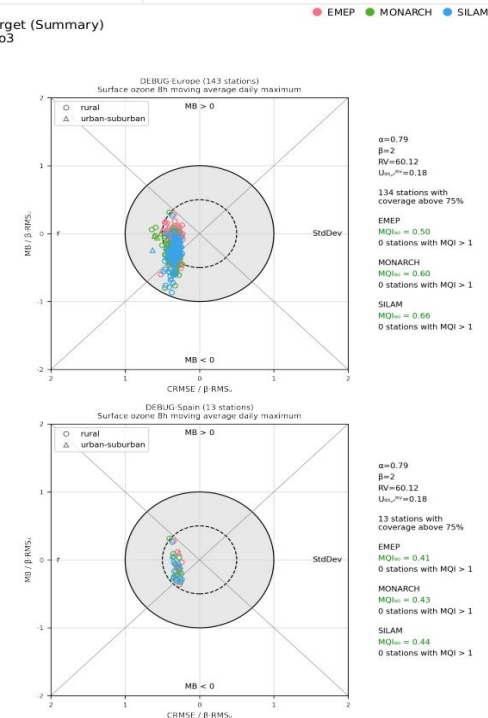
Violin (Summary)
EBAS[sconco3]



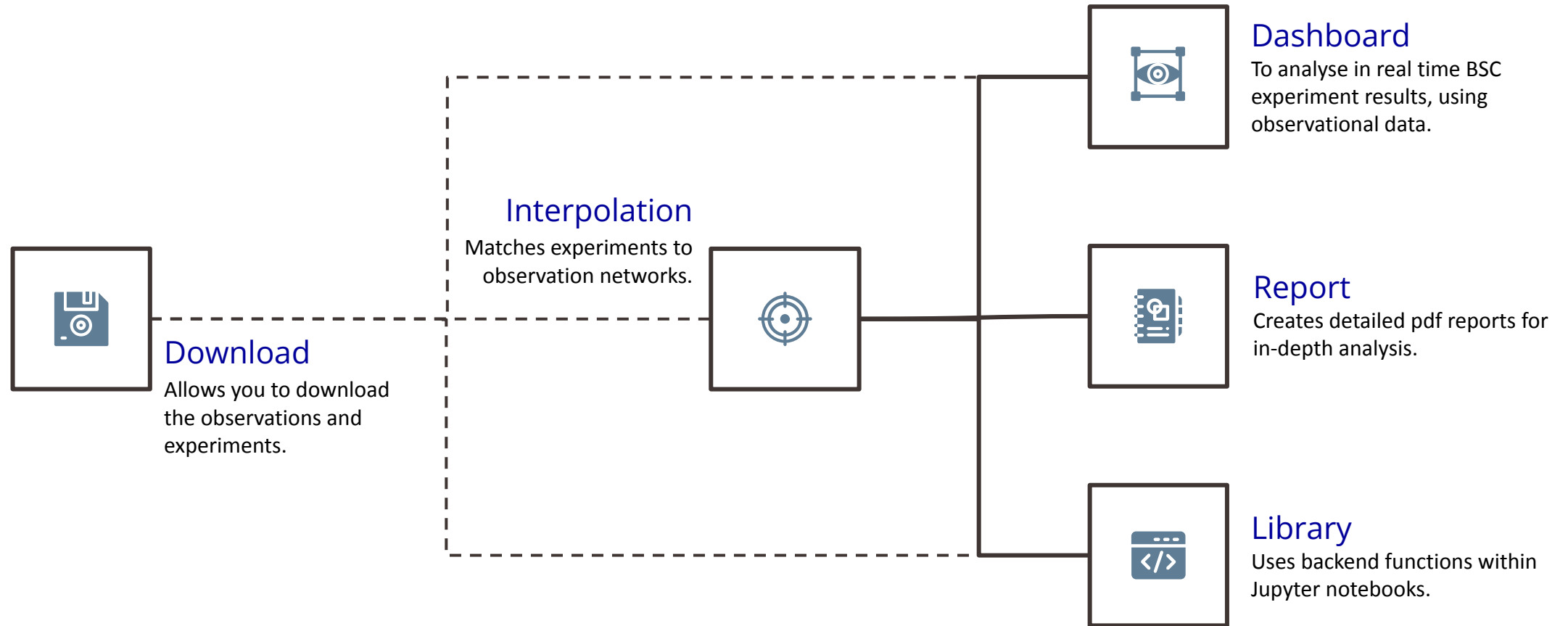
Timeseries (Summary)
EBAS[sconco3]



Fairmode target (Summary)
EBAS[sconco3]



Structure



Library

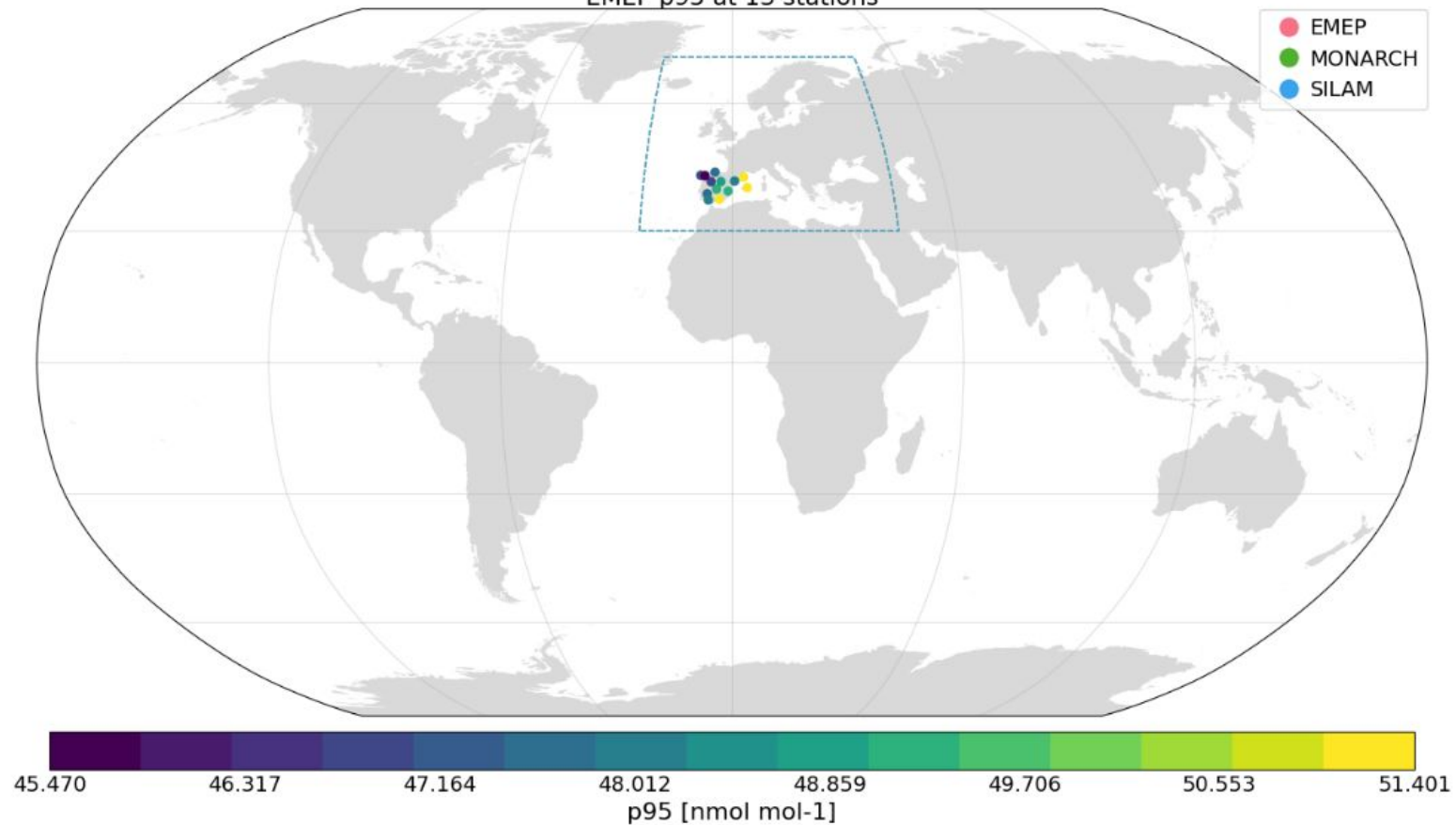
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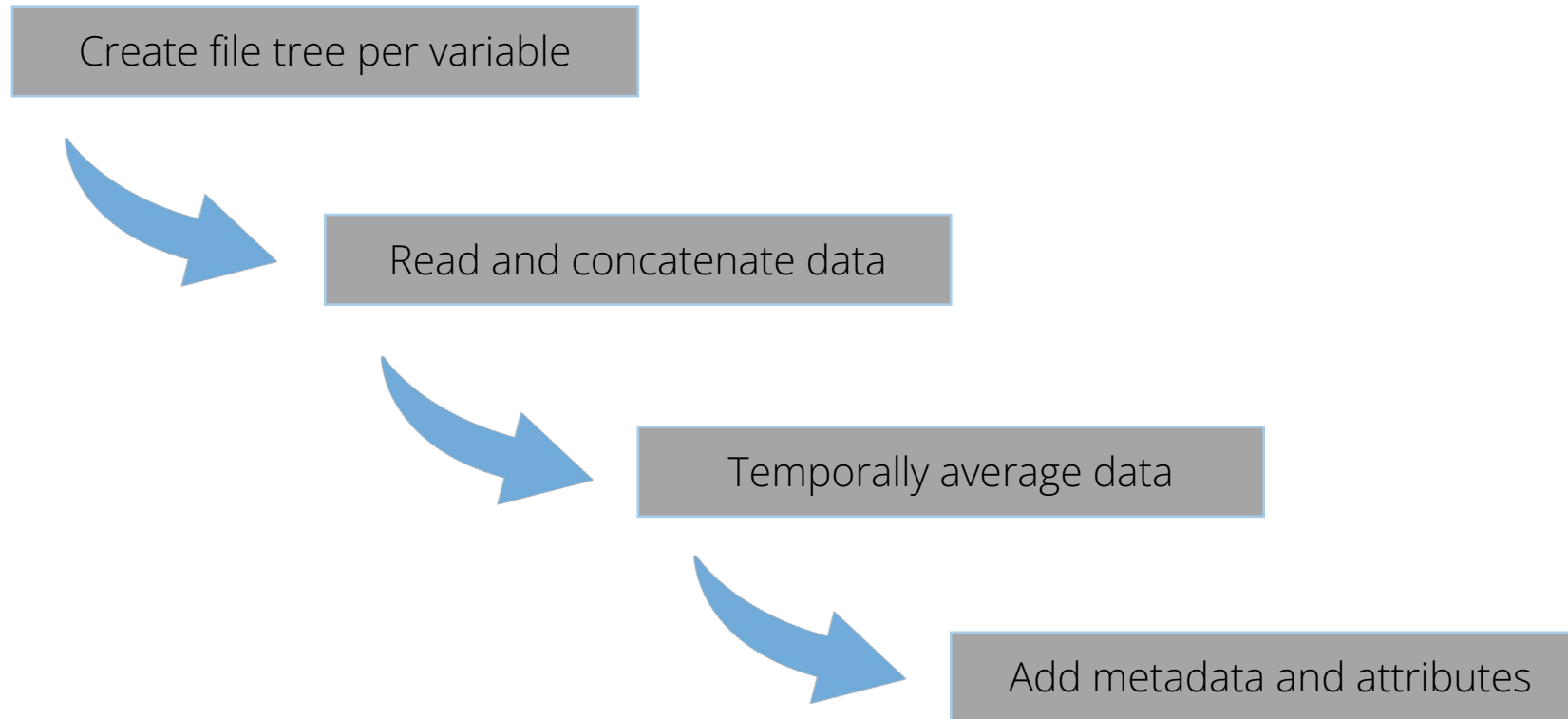
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Earth Sciences
Department



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